

DİDEM ARSLAN YENİHAYAT

Data Scientist · Former R&D Leader, Chemical Engineering

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PROFILE

Chemical engineer and R&D leader with 10+ years diagnosing complex industrial systems — now applying that same pattern-first approach to data. Co-inventor on five patents and former manager of a 10-person R&D team, with a track record of turning fragmented operational data (formulation, quality, production) into root-cause findings and process decisions. Currently completing a Data Science & Analytics Bootcamp (Workintech), building Python, machine learning and SQL/data-pipeline fluency to formalize a skill set developed informally over a decade of R&D problem-solving.

DATA & ANALYTICS SKILLS

Data engineering & pipelines: SQL (BigQuery dialect), BigQuery, dbt, Fivetran, Git, Wikidata SPARQL.

Programming & ML: Python (pandas, numpy, scikit-learn, matplotlib, seaborn); linear & log-transformed regression, model evaluation (R^2 , MAE).

Statistical methods: hypothesis testing (Mann-Whitney U), Pearson correlation, time-series autocorrelation, descriptive statistics, robustness/outlier checks.

Visualization & reporting: Power BI, Looker Studio (Google Data Studio), Google Sheets.

Automation: Zapier.

(Workintech Data Science & Analytics Bootcamp, 2026)

Analytical foundation: root-cause analysis, statistical process reasoning, competitive/benchmarking analysis, structured problem framing (Stage-Gate / Agile), technical reporting for cross-functional and non-technical audiences.

FEATURED DATA PROJECT

The Recognition Lag — Who Gets Recognized, and When? Analysis of institutional recognition patterns across 160,699 MoMA works. Applied Mann-Whitney U and Pearson correlation tests to quantify acquisition-lag gaps by gender and geography, and lag-1 autocorrelation to test temporal consistency in acquisition pace. Built a log-transformed regression model estimating artists' unrealized productivity ("lost potential") from early-career output ($R^2=0.705$, $n=5,656$), validated against known cases (e.g., Keith Haring: actual 49 works vs. predicted 48.8).

Full write-up, methodology notes, and code: [portfolio site URL]

In progress: additional data-science portfolio projects (Dose of Reality, Reading the Green Line, Brick by Brick, Multiverses: Character Survival Analysis) — added as each is completed.

PROFESSIONAL EXPERIENCE

R&D Specialist → Executive → Manager, Coatings & Additives

Jul 2015 – Jul 2025

EGE Kimya San. ve Tic. A.Ş. · Sakarya, Türkiye

- Progressed over 10 years from formulation R&D to leading a 10-person team; introduced a structured Agile / Stage-Gate methodology to improve project throughput and decision governance.
- Acted as the analytical focal point between R&D, Production, Quality and Sales — running root-cause analysis on off-spec production data and driving corrective action.
- Led a sustainability initiative (Zero Liquid Discharge + circular-system design) that cut process waste by over 80% while recovering reusable chemicals, based on process-data analysis.
- Developed and commercialized 50+ coatings additives; ran competitive benchmarking studies to inform product strategy.

EDUCATION

Data Science & Analytics Bootcamp (in progress) — Workintech · Jun 2026 – present

BEng-equivalent, Chemical Engineering — Hacettepe University, Ankara, Türkiye · 2009 – 2014

PATENTS & PUBLICATIONS

Co-inventor on 5 patents / patent applications in coatings additives, driers and polyurethane catalysts (EGE Kimya, 2019–2024). Author, "Cobalt Octoate and MEKO — An Eco-friendly and Sustainable Alternative: EGEDry Alpha," Turkcoat Magazine, 2017 (co-author). Full list available on request or via portfolio site.

CERTIFICATIONS & LANGUAGES

ISO 14064 GHG Verifier — NBC Cert (2025–2030) · Negotiation Mastery — Harvard Business School (2022) · Project, Change & Risk Management, Leadership — IBMI (2022)

Claude 101 — Anthropic · Google AI Essentials — Google

Languages: English — Full Professional Proficiency · Turkish — Native · German — Elementary

REFERENCES

Available upon request.